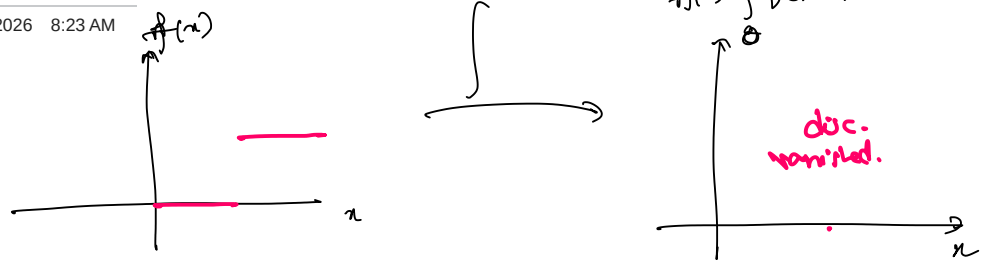
Lecture 19

Monday, 21 September 2026 8:23 AM

Recap:



FTC: If f is continuous on $[a,b]$ then

$$A(x) = \int_a^x f(t)\,dt \quad \text{is} \quad \underline{\text{diff'ble}} \quad \text{on} \quad \underline{[a,b]}$$

SFTC: If $P'(x) = f(x)$ then $\int_a^b f(x)\,dx = \underline{P(b) - P(a)}$

P is called $\underline{\text{Primitive}}$ of f

Today: $\dfrac{d}{dx}\left(\displaystyle\int_{a(x)}^{b(x)} f(t)\,dt\right) = ?$

Let P be a primitive of f

ie $P'(x) = f(x)$

$$\frac{d}{dx}\left(\int_{a(x)}^{b(x)} f(t)\,dt\right) = \frac{d}{dx}\left(\int_{a(x)}^{b(x)} P'(t)\,dt\right)$$

$$\overset{\text{SFTC}}{=} \frac{d}{dx}\left(P(b(x)) - P(a(x))\right)$$

$$\overset{\text{chain rule}}{=} P'(b(x)).b'(x) - P'(a(x)).a'(x)$$

$$\boxed{\frac{d}{dx}\left(\int_{a(x)}^{b(x)} f(t)\,dt\right) = f(b(x)).b'(x) - f(a(x)).a'(x)}$$

Eg1: Let $f(x) = \displaystyle\int_x^2 \frac{1}{\sqrt{1+t^3}}\,dt$. $x > 0$

2 x 1

Find $\int_0^2 x\, f(x)\, dx$.

$f'(x) =$

$\int_a^a g(x)\, dx = 0$

$$\frac{d}{dx} f(x) = \frac{d}{dx}\left(\int_x^2 \frac{1}{\sqrt{1+t^3}}\, dt \right)$$

Formula

$$= \frac{1}{\sqrt{1+2^3}} \cdot \frac{d}{dx}(2) - \frac{1}{\sqrt{1+x^3}} \cdot \frac{d}{dx}(x)$$

$$\frac{d}{dx} f(x) = -\frac{1}{\sqrt{1+x^3}}$$

$$f'(x) = -\frac{1}{\sqrt{1+x^3}}$$

$$\int_0^2 x\, f(x)\, dx = \int_0^2 \underline{f(x) \cdot x}\, dx =$$

Int. by parts

$$= f(x) \cdot \frac{x^2}{2}\Big|_{x=0}^{2} - \int_{x=0}^{2} f'(x) \cdot \frac{x^2}{2}\, dx$$

$$= \left[f(2) \cdot \frac{2^2}{2} - f(0) \cdot \frac{0^2}{2} \right] - \int_{x=0}^{2} \frac{-1}{\sqrt{1+x^3}} \cdot \frac{x^2}{2}\, dx$$

$f(2) = ?$, $f(0) = 0$

$$= \int_{x=0}^{2} \frac{x^2}{2\sqrt{1+x^3}}\, dx = \boxed{\frac{2}{3}} \quad ?$$

check!

Q: Connection between Derivatives of f and f^{-1}: ?

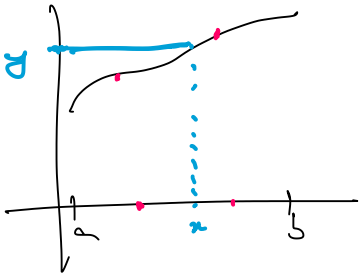
f, f' continuous on $[a, b]$ and

Thm: Let f be [illegible] strictly increasing on $[a,b]$. Then f is invertible.

Let $g = f^{-1}$.

If f is diff'ble at $x=p$ and $f'(p) \neq 0$ then g is diff'ble at $f(p)$

and $$g'(f(p)) = \frac{1}{f'(p)}.$$



$y = f(x)$

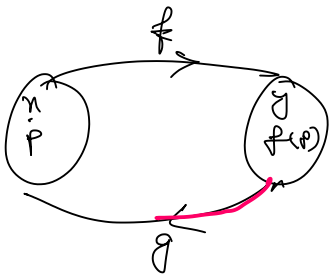
$x = f^{-1}(y)$

OR

$x = g(y)$

ie. $$g'(y) = \frac{1}{f'(x)}$$

ie $$\boxed{\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}}$$

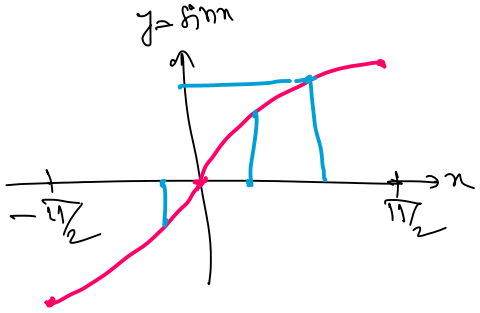


Eg:1) Let $y = \sin x$ $\quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Find $\frac{dx}{dy} = ?$ (in term of y).

$y = \sin x$ is invertible as sin is Cont & strictly ↑

$\frac{dy}{dx} = \cos x > 0 \quad x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$



$$\frac{dx}{dy} \overset{\text{Thm}}{=} \frac{1}{\left(\frac{dy}{dx}\right)}$$

$$\frac{dx}{dy} = \frac{1}{\cos x} = \frac{1}{\sqrt{1-y^2}}$$

$$\frac{d}{dy}\left(\sin^{-1} y\right) = \frac{1}{\sqrt{1-y^2}}$$

$y'' = -\sin x > 0$ $\quad x \in \left(\frac{-\pi}{2}, 0\right)$

$y \in (-1,1)$

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Eg2): Let $f(x) = \int_{x}^{} \frac{1}{\quad} dt.$

$\int_0^{\cdot} \sqrt{1+t^3}$

(i) Show that f is invertible. Let $g = f^{-1}$.

(ii) Prove that $g''(y) = \frac{3}{2} g^2(y)$.

$$f'(x) = \frac{1}{\sqrt{1+x^3}} > 0$$

$\therefore$ f is cont & strictly $\uparrow$

$\xrightarrow{\text{Thm}}$ f is invertible.

Let $g = f^{-1}$

$y = f(x)$
$x = g(y)$

$$g'(y) \overset{\text{Thm}}{=} \frac{1}{f'(x)} = \frac{1}{\frac{1}{\sqrt{1+x^3}}} = \sqrt{1+x^3}$$

(x = function of y here)

$\frac{dx}{dy}$

$$g''(y) = \frac{1}{2\sqrt{1+x^3}} \cdot 3x^2 \cdot \frac{dx}{dy}$$

$$= \frac{3}{2} \cdot \frac{x^2}{\sqrt{1+x^3}} \cdot \sqrt{1+x^3}$$

$$g''(y) = \frac{3}{2} x^2 = \frac{3}{2} \cdot (g(y))^2$$